

MATH3881/5881 — Statistical Machine Learning Theory

Sample Short-Answer Questions (Mid-Term Exam, Term 2 2026)

Purpose. These four questions illustrate the style and topic breadth of the short-answer section of the mid-term exam. They are *not* the actual exam questions. The actual mid-term will contain four short-answer questions of comparable style and total mark value (60 marks; 15 marks each), covering material up to Week 5.

Closed-book exam. The mid-term is closed-book. Any formulas or definitions from the lecture notes that a question relies on will be provided in a Support Material along with the paper, as done below.

Support Material

ReLU: $\text{ReLU}(z) = \max(0, z)$, applied entry-wise to vectors and matrices.

Universal Approximation Theorem (one-dimensional form): for every continuous $f^* : [a, b] \rightarrow \mathbb{R}$ and every $\varepsilon > 0$, there is a one-hidden-layer network f_θ with a non-polynomial activation such that $|f_\theta(x) - f^*(x)| < \varepsilon$ for all $x \in [a, b]$.

Adam optimiser (scalar parameter θ , gradient $g^{(t)}$ at iteration t): with $v^{(0)} = s^{(0)} = 0$ and smoothing parameters $\beta, \gamma \in [0, 1)$,

$$\begin{aligned} v^{(t+1)} &= \beta v^{(t)} + (1 - \beta) g^{(t)}, & \hat{v}^{(t+1)} &= \frac{v^{(t+1)}}{1 - \beta^{t+1}}, \\ s^{(t+1)} &= \gamma s^{(t)} + (1 - \gamma) (g^{(t)})^2, & \hat{s}^{(t+1)} &= \frac{s^{(t+1)}}{1 - \gamma^{t+1}}, \end{aligned}$$

and the parameter update is

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \frac{\hat{v}^{(t+1)}}{\sqrt{\hat{s}^{(t+1)} + \varepsilon}}.$$

Sample S1. Pooling and ReLU: swap or not?

For a window $x \in \mathbb{R}^{2 \times 2}$, define

$$\text{maxpool}(x) = \max_{i,j} x_{ij}, \quad \text{avgpool}(x) = \frac{1}{4} \sum_{i,j} x_{ij}.$$

(a) Prove that

$$\text{ReLU}(\text{maxpool}(x)) = \text{maxpool}(\text{ReLU}(x)) \quad \text{for every } x \in \mathbb{R}^{2 \times 2}.$$

[5 marks]

(b) Does the same identity hold for average pooling? Prove it or give a counterexample. [5 marks]

(c) A CNN applies ReLU and then a 2×2 , stride-2 max-pool. By (a), the two operations can be swapped without changing the network's output. Which order is computationally cheaper, and by what factor? [5 marks]

Answer S1

Answer S1 (Continue)

Sample S2. Reverse-mode differentiation on a computational graph.

Let

$$C(\theta_1, \theta_2) = (\theta_1 + \theta_2^2) e^{\theta_2^2},$$

evaluated at $(\theta_1, \theta_2) = (1, 1)$. The *adjoint* of a node a in a computational graph is $\zeta_a := \partial C / \partial a$.

- (a) Decompose C into elementary operations (+, $\times$, squaring, exp) and draw the computational graph. Identify the node with two outgoing edges. [4 marks]
- (b) Forward pass: compute the value at every node. [3 marks]
- (c) Prove: if a node a feeds exactly the nodes $u_1, \dots, u_m$, then

$$\zeta_a = \sum_{j=1}^m \zeta_{u_j} \frac{\partial u_j}{\partial a}.$$

[4 marks]

- (d) Run the reverse pass to obtain $\nabla C(1, 1)$. Verify $\partial C / \partial \theta_2$ by direct differentiation. [4 marks]

Answer S2

Answer S2 (Continue)

Sample S3. Exact representation by shallow ReLU networks.

A one-hidden-layer ReLU network on $\mathbb{R}$ is a function of the form

$$f_{\theta}(x) = \sum_{i=1}^N \alpha_i \text{ReLU}(w_i x + b_i) + c, \quad \alpha_i, w_i, b_i, c \in \mathbb{R}.$$

- (a) Find such an f_{θ} with $N = 2$ satisfying $f_{\theta}(x) = |x|$ for every $x \in \mathbb{R}$, and prove the identity. [4 marks]
- (b) Prove that no such f_{θ} with $N = 1$ satisfies $f_{\theta}(x) = |x|$ for every $x \in \mathbb{R}$. [4 marks]
- (c) Prove that every f_{θ} with N units is piecewise-linear with at most $N + 1$ linear pieces. [4 marks]
- (d) Deduce that no one-hidden-layer ReLU network satisfies $f_{\theta}(x) = x^2$ for every $x \in [0, 1]$, no matter how large N is. In one sentence, explain why this does not contradict the Universal Approximation Theorem. [3 marks]

Answer S3

Answer S3 (Continue)

Sample S4. Adam on an intermittent gradient.

Consider the Adam optimiser (see the Support Material) applied to a single scalar parameter θ . At every iteration $t = 1, 2, 3, \dots$ the gradient $g^{(t)}$ is drawn independently: $g^{(t)} = g$ with probability q and $g^{(t)} = 0$ with probability $1 - q$, where $g > 0$ and $q \in (0, 1]$ are fixed.

- (a) Let $u^{(t+1)} = \beta u^{(t)} + (1 - \beta) \xi^{(t)}$ with $u^{(0)} = 0$ and $\xi^{(t)}$ iid with mean m . Prove that $\mathbb{E}[u^{(t+1)}] = (1 - \beta^{t+1})m$ for all $t \geq 0$, and hence $\mathbb{E}[u^{(t+1)}] \rightarrow m$ as $t \rightarrow \infty$. [4 marks]
- (b) Compute $\lim_{t \rightarrow \infty} \mathbb{E}[v^{(t)}]$ and $\lim_{t \rightarrow \infty} \mathbb{E}[s^{(t)}]$ in terms of g and q . [3 marks]
- (c) For large t the bias-correction factors are ≈ 1 ; also set $\varepsilon = 0$ for simplicity. Approximating the step by replacing $\hat{v}$ and $\hat{s}$ with their limiting means, show that $|\theta^{(t+1)} - \theta^{(t)}| \approx \alpha \sqrt{q}$. [4 marks]
- (d) Standard gradient descent with learning rate α takes the step $-\alpha g^{(t)}$ at each iteration. Show that its average step magnitude is $\alpha g q$. Using (c), compute the ratio

$$\frac{\text{Adam step magnitude}}{\text{average gradient-descent step magnitude}},$$

and state its limit as $q \rightarrow 0$.

[4 marks]

Answer S4

Answer S4 (Continue)